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Onsite photovoltaic and energy storage deployment in grid-interactive data centers: Maximizing renewable utilization and profitability under lifecycle progressive loading

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ABSTRACT

In the era of rapidly growing AI demand, the deployment of onsite photovoltaic (PV) and electrical energy storage (EES) represents a significant opportunity for data center decarbonization. This study investigated the dual benefits of renewable energy utilization and lifecycle profitability in grid-interactive data centers with onsite PV and EES deployment. Specifically, the optimal configurations of onsite PV and EES capacity are determined under lifecycle progressive loading and two representative electricity markets, using an hourly-resolution EES dispatch optimization algorithm for maximizing renewable utilization and profitability. We found that grid-interactive data centers can achieve significantly high renewable energy utilization, while still delivering strong economic returns over their lifetime, despite the high upfront capital investment required for onsite PV and EES deployment. Specifically, a 100 kW data center with 120 kW PV and 100 kWh EES could achieve a renewable utilization of 50-51% in both markets, yielding lifecycle net profits of $0.26 million in Shenzhen market and $0.04 million in CAISO market. Achieving over 90% renewable utilization while maximizing lifecycle net profit requires substantial capacity expansion: a 2.2-fold increase in EES and an 8-fold increase in PV in Shenzhen, and a 1.4-fold increase in EES and a 10-fold increase in PV in CAISO. Under these conditions, lifecycle profit remains $0.04 million for Shenzhen, but becomes negative for CAISO.

Nomenclature

AI	Artificial Intelligence
HPC	High-Performance Computing
PUE	Power Usage Effectiveness
IT	Information Technology
PV	Photovoltaic
EES	Electrical Energy Storage
UPS	Uninterruptible Power Supply
CAISO	California Independent System Operator
STC	Standard Test Condition
ToU	Time-of-Use
RE	Renewable Energy
SOC	State of Charge
E	Energy (kWh)

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P	Rated Power (kW)
G	Solar Irradiance (W/m^2)
A	Area (m^2)
C	Cost ($)
fr	Frequency Regulation (kW)
r	Frequency Regulation Revenue ($)
π	Electricity Price ($)
λ	Performance Ratio
REV	Revenue
Subscripts	
min	Minimum
max	Maximum
init	Initial state
final	Final state

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(*continued*)

t	At time *t*
dem	Demand
use	Used power
gen	Generated power
cur	Curtailment
uti	Utilization
sur	Surplus
def	Deficit
inv	Investment
rep	Replacement
tot	Total
i	At ith year
d	Discount Rate
c	Cycle
O&M	Maintenance
base	Baseline Scenario

1. Introduction

The worldwide growth of generative AI and high-performance computing is driving the rapid development of data centers [1]. Although these facilities support the digital economy, they also require extremely large amounts of electricity and produce substantial carbon emissions, creating increasing environmental pressure [2,3]. In response to climate targets, the adoption of renewable energy sources, including solar and wind power, has become an important approach for reducing the environmental impact of data centers [4].

1.1. Opportunities and challenges for renewable-powered data centers

To lower the environmental impact of data centers, renewable energy technologies such as solar photovoltaic (PV) systems [5] and wind turbines are being adopted more widely [6]. Among these technologies, onsite renewable generation, particularly onsite PV deployment, has become an increasingly common strategy for decarbonizing buildings [7,8] and data center facilities [9]. Generating electricity directly where it is consumed helps data centers decrease their dependence on conventional power grids [10]. Several major IT companies have already applied this approach in practice. For instance, Apple's data center in North Carolina uses multiple renewable energy sources and produces about 244 million kWh of clean electricity annually [11]. Similarly, Google's Belgium data center includes onsite solar installations capable of generating as much as 2.9 GW-hours of renewable electricity per year [12].

Despite these advantages, onsite PV systems face an important limitation due to the intermittent nature of solar energy [13]. Because solar radiation changes over the course of the day, electricity generation is high during periods such as midday but drops significantly at night. To address this issue, electrical energy storage (EES) systems can store excess renewable electricity produced during peak solar periods [14]. The stored electricity may later be released when solar generation is insufficient, helping maintain a stable and reliable renewable energy supply for data center operation [15]. In addition, combining PV systems with EES systems can further reduce grid dependence, lower electricity expenses, and improve the sustainability performance of data centers.

Combining renewable energy systems with electrical energy storage not only enhances the sustainability performance of data centers, but also contributes to grid flexibility and stable grid operation [16]. Through interaction with the power grid, data centers can support ancillary services, including frequency regulation [17], which plays an important role in maintaining grid stability. In return, utilities and electricity markets can offer financial compensation to data centers for providing these services [18]. In addition, battery systems enable energy arbitrage strategies, where electricity is stored during periods of low power prices and discharged when electricity prices are high, thereby reducing operational costs [19]. Several studies have investigated these benefits. For example, Zhang et al. [20] examined how surplus battery storage capacity in data centers can be utilized to provide grid flexibility services while minimizing lifecycle electricity costs. Their results showed considerable economic advantages, with the battery system achieving returns equal to 1.29 times the initial investment. Mamun et al. [21] developed a health-aware optimal control framework for lithium-ion batteries in data center demand response applications. Their study formulated a multi-objective optimization approach that simultaneously considers electricity cost savings and battery degradation.

1.2. Literature review on renewable energy integration in data centers

The integration and effective utilization of renewable energy in data centers have attracted increasing research attention. Current studies mainly explore this issue from two perspectives: (1) operational methods, such as IT workload scheduling and the application of energy storage systems; and (2) capacity sizing optimization of renewable energy and storage systems during the design stage.

1.2.1. Renewable-adaptive IT workload scheduling

Grange et al. [22] developed a renewable-energy-aware scheduling method for batch workloads in data centers that increases the use of onsite solar generation and reduces dependence on grid electricity. Compared with conventional scheduling approaches, their method lowered brown energy consumption by as much as 49% and reduced operating costs by up to 51%. Rostirolla [23] further examined scheduling strategies for data centers supplied entirely by onsite renewable energy, where workloads are adjusted according to fluctuations in renewable power availability to improve both service quality and profitability. The proposed meta-heuristic algorithms achieved energy savings of up to 14.12% while decreasing task violations from 188 to 11 under limited renewable energy conditions.

1.2.2. Energy storage integration for renewable utilization

Li et al. [24] explored two strategies, namely opportunistic workload scheduling and battery energy storage, to improve renewable energy utilization in small- and medium-scale data centers using real workload and solar generation data. Their findings showed that the combined use of both approaches reduced brown energy consumption by as much as 33% compared with using only battery storage systems. Peng et al. [25] developed an energy management framework that combines onsite renewable generation with coordinated control of distributed UPS systems to lower overall data center electricity costs. Acun et al. [26] investigated the use of grid-scale renewable energy together with lithium-ion battery systems to support 24/7 carbon-free operation in hyperscale data centers. Their study indicated that current electrical energy technologies are capable of delivering 28 MW for 4 h, which matches typical hyperscale data center power demands ranging from 20 to 40 MW.

1.2.3. Sizing optimization for renewable systems and storage

Cupelli et al. [27] developed an optimization method for sizing battery energy storage systems in data centers to support frequency containment reserve services. Their analysis showed that a 560 kWh lithium-ion battery system could lower operating costs by around 4%. Li et al. [28] introduced a hybrid optimization approach that combines opportunistic workload scheduling with electrical energy storage sizing to improve renewable energy utilization in data centers. The study found that integrating optimal battery sizing with time-shifted workload management decreased brown energy consumption by 33% compared with approaches relying only on energy storage. Figini et al. [29] proposed a scenario-based optimization framework for the joint sizing of PV systems and EES in data centers while considering both carbon emissions and operating costs. Their results revealed strong regional differences and emphasized the importance of local grid carbon intensity and solar irradiance conditions in determining optimal system sizes.

1.3. Main innovations and contributions

Previous research has mainly concentrated on a single dimension of data center energy management, such as short-term operational control or fixed system sizing. However, few studies have been given to integrated frameworks that simultaneously optimize both the system design and operational strategy of onsite PV and EES systems, particularly under progressive lifecycle load growth and participation in electricity markets. The main contributions of this study are summarized as follows.

First, hourly-resolution EES dispatch in grid-interactive data centers is optimized under specific IT loading conditions, PV variation and electricity markets. Second, the optimal configurations of onsite PV and EES capacity are determined based on the daily optimal dispatch algorithm, accounting for lifecycle progressive IT loading and different optimization prioritization. Third, a comparative analysis of optimal PV-EES configurations is conducted across two electricity markets, Shenzhen and CAISO, to examine the trade-off between maximizing lifecycle profitability and renewable energy utilization. Furthermore, the impacts of PV-EES configurations and objective prioritization on EES utilization and PV curtailment are also discussed. The results provide critical guidance for decarbonizing emerging AI data centers and offer actionable insights for policymakers and industry stakeholders to support future planning and strategic decision-making.

2. Data centers integrated with onsite photovoltaic and energy storage systems

2.1. Data centers with PV-EES integration

Fig. 1 presents the configuration of the integrated PV-EES system together with the major energy flow pathways within the system. This configuration provides multiple benefits, including reduced dependence on the conventional power grid, lower carbon emissions, possible reductions in electricity costs, and improved grid stability [20].

2.2. Reference data center configuration

The reference data center used in this study is described in Ref. [30], with its key specifications summarized in Table 1. The facility is designed to handle a peak IT load of 100 kW. To improve energy efficiency and reduce dependence on grid electricity, the data center is equipped with a PV system and an EES unit. The PV array has a rated capacity of 150 kW under standard test conditions, which allows it to cover the full IT load under optimal solar irradiance. Complementing the PV system is an EES unit with a storage capacity of 500 kWh, which serves to buffer intermittent solar generation and improve overall energy reliability.

The energy management strategy is based on a prioritized dispatch sequence. During operation, electricity generated by the PV system is used first to supply the data center load whenever solar power is available. When PV generation cannot fully satisfy the IT power demand, additional electricity is supplied by the EES if sufficient stored energy remains available. The utility grid is used only when both the PV system and the EES are unable to completely meet the load requirement.

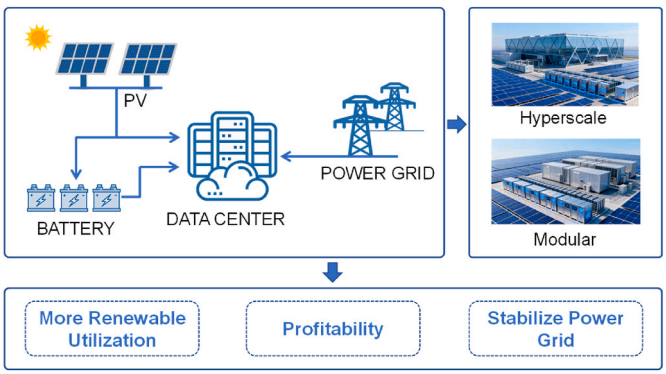


Fig. 1. Schematic of the integrated PV-EES data centers.

Table 1
Parameters for PV-EES integrated data centers.

Parameter	Value
Design IT load	100 kW [30]
Data center lifespan	20 years [31]
Design capacity of EES	500 kWh [30]
Energy storage efficiency	0.995 [32]
Design PV array	150 kW [30]
PV tilt angle	Local latitude [33]
Performance ratio	0.8 [34]
Capacity cost of PV	500 $/kW [35]
Capacity cost of EES	200 $/kWh [36]
O&M cost/year of EES	0.5% of investment cost [37]
Battery cycle (to end-of-life)	9000 [38]
Battery float lifespan	10 years [39]
Charging/discharging efficiency	0.95 [38]
Energy-to-Power ratio (γ)	2.8 [38]
Discount rate (d)	4% [40]

2.3. Progressive IT loading and PV power generation

The operational dynamics of a data center with integrated PV and EES systems are significantly influenced by three key factors: (1) the progressive increase in IT load over time, and (2) the inherent temporal variability of solar power generation. (3) the capacity of the EES system. The EES plays a critical role in mitigating PV intermittency and ensuring a continuous power supply to the IT load. Understanding the interaction between them is essential for designing effective energy management strategies.

2.3.1. Progressive IT load growth

Data center IT loads typically do not operate at full design capacity from the outset. Instead, they follow a staged or progressive deployment model, where computing resources and associated power demand gradually increase over time, as illustrated in Fig. 2. This staged growth is often driven by business expansion, phased server installations, or demand scaling [41]. It is assumed that there are five stages in total. Stage 1 to 5 (s1∼s5) IT loads represent 20% to 60% of the design load, respectively [42].

2.3.2. PV power generation

Compared with the relatively stable and controllable characteristics

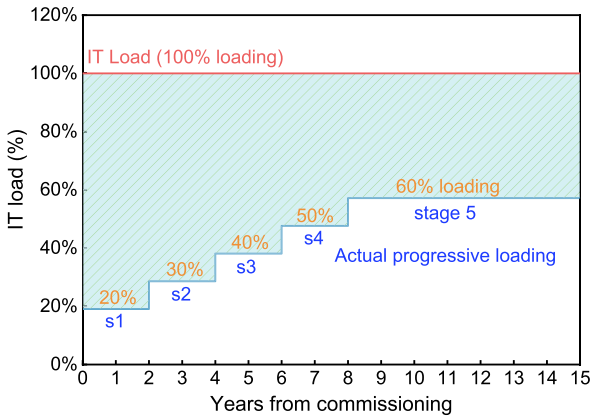


Fig. 2. Typical progressive IT Load Growth throughout the lifetime.

of IT load demand, solar power generation varies significantly over time because solar irradiance is strongly affected by weather conditions, seasonal variation, and day–night cycles. Such fluctuations create challenges for maintaining a stable and reliable electricity supply, especially when renewable energy is prioritized in data center operations.

The actual hourly electricity generation of the PV system is determined by local solar irradiance conditions together with system efficiency parameters. Estimating PV power generation under changing weather conditions on an hourly basis is essential for energy dispatch and operational control within the integrated energy management system. The hourly PV electricity output is calculated using Equation (1). The corresponding solar irradiance data were sourced from Ref. [43].

$$P_{PV,t}^{gen} = P_{rated} \times \left(\frac{G_t}{G_{STC}}\right) \times \lambda \quad (1)$$

where, $P_{PV,t}^{gen}$ denotes actual power output at time t (kW); P_{rated} denotes the rated power of the PV array (kW); G_t represents the average solar irradiance at time t (W/m^2); G_{STC} is the standard test condition irradiance, taken as 1000 W/m^2; The performance ratio (λ), a dimensionless coefficient, taken as 0.8 in this study [34], accounts for losses due to factors such as temperature, soiling, and inefficiencies in inverters and wiring.

2.3.3. Operational case based on PV-IT load matching

To systematically analyze system performance under varying conditions of PV generation and IT load demand, five distinct operational scenarios are defined based on the relative balance between PV energy surplus and PV energy deficit over a defined analysis period (e.g., daily or hourly basis). These scenarios are graphically represented using the areas under the PV generation and IT load curves, referred to as A_{sur} (PV surplus) and A_{def} (PV deficit), respectively.

Case 0. No PV Surplus ($A_{sur} = 0$)

This case indicates that, throughout the considered period, PV generation is insufficient to meet the IT load at any time. In this scenario, the system relies entirely on the EES (if available) and/or the grid for power supply.

Case 1. PV Surplus Less Than PV Deficit ($A_{sur} < A_{def}$)

In this scenario, PV generation alone is insufficient to meet the full energy demand, even with energy shifting via storage (Fig. 3A). It is also divided into two subcases:

Subcase 1.1: Sufficient Storage Capacity

Although the total PV energy is insufficient, the EES can at least shift all available surplus energy to the deficit periods, minimizing grid reliance as much as possible.

Subcase 1.2: Insufficient Storage Capacity

Both PV generation and storage capacity are insufficient to cover the IT load, resulting in a significant portion of the demand being met by grid electricity.

Case 2. PV Surplus Greater Than or Equal to PV Deficit ($A_{sur} \geq A_{def}$)

In this scenario, the total PV energy generated exceeds or matches the total energy deficit (Fig. 3B). This case is further divided into two subcases:

Subcase 1.1: Sufficient Storage Capacity

The energy storage system has enough capacity to store the surplus PV energy and subsequently discharge it to fully cover the energy deficit periods. In this ideal condition, grid dependency can be entirely avoided.

Subcase 1.2: Insufficient Storage Capacity

Although the total PV energy is theoretically adequate, the EES capacity is not sufficient to store all the surplus energy. As a result, some surplus PV energy is curtailed, and additional energy must be drawn from the grid during deficit periods.

These operational cases provide a framework for evaluating the effectiveness of PV-EES systems in real-world scenarios with dynamic load profiles and variable renewable energy inputs. The classification also aids in identifying opportunities for system optimization, such as increasing storage capacity, implementing demand-side management, or integrating predictive control algorithms for energy dispatch.

2.4. Representative electricity markets and data center energy flexibility

Electricity pricing mechanisms vary by region and are largely determined by the market structures established by independent system operators. For instance, the California Independent System Operator (CAISO) in the United States operates a combined energy and ancillary service market, allowing electricity and ancillary services to be traded simultaneously. By comparison, China's electricity market commonly applies Time-of-Use (ToU) pricing, where electricity rates differ during peak, flat, and off-peak periods to encourage load shifting and alleviate peak grid demand. In this study, two representative electricity markets, namely CAISO in the United States and the ToU pricing market in Shenzhen, China, are investigated to evaluate the optimal design of renewable-powered data centers integrated with energy storage systems. Historical data for 2023 from the CAISO energy markets was obtained from the official website [44], and data for the Shenzhen ToU market was obtained from Refs. [45,46].

In data centers, this flexibility can be utilized for both energy arbitrage and ancillary grid services, including frequency regulation. Energy arbitrage reduces electricity expenses by shifting power consumption from periods with high electricity prices to periods with lower prices. Ancillary services, on the other hand, require fast-response operation capabilities. For example, frequency regulation depends on automatic

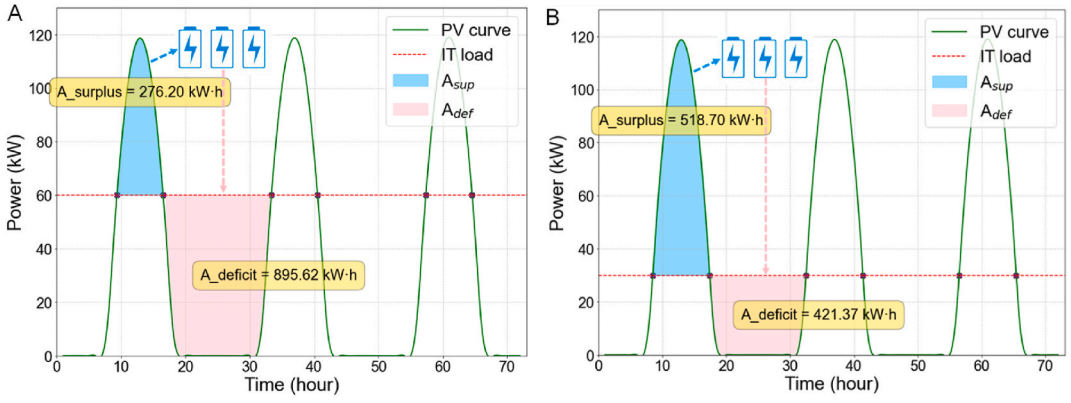


Fig. 3. Schematic of operational cases based on PV-IT load matching.

generation control signals dispatched at second-level timescales to correct deviations in grid frequency. By participating in these electricity markets, demand-side resources can help support grid reliability while also generating additional economic benefits.

3. Optimization framework for Photovoltaic and electrical energy storage integrated system design

This section introduces an optimization framework aimed at enhancing economic gains from energy flexibility services while maintaining the reliable operation of a data center facing gradually increasing IT loads and variable PV generation. The framework combines optimal dispatch strategies for the PV-EES system with consideration of technical limits and opportunities for market participation. Its primary objectives are to maximize the use of renewable energy and jointly optimize energy cost savings.

3.1. Formulation of the optimal dispatch model

3.1.1. Objective function

The primary objective of this study is to maximize PV self-consumption by utilizing an energy storage system to store surplus renewable energy generated by PV panels and release it when renewable energy production is insufficient. This objective can be formulated as:

$$\max \frac{\sum_{t=1}^{T} P_{PV,t}^{use}}{\sum_{t=1}^{T} P_{dem,t}} \tag{2}$$

where, $P_{PV,t}^{use}$ is the used PV power at time t; $P_{dem,t}$ is the power demand of the data center at time t.

On the basis of maximizing PV self-consumption, a secondary objective is introduced: to maximize the net operational profit over a defined time horizon. The objective function incorporates revenue from flexibility markets, cost savings resulting from PV self-consumption, and reduced expenditures from avoided grid electricity purchases. Mathematically, the profit maximization objective is expressed as:

$$\max \left\{ \sum_{t=1}^{T} \left(r_t \cdot fr_t - \pi_t \cdot P_{EES,t} \right) \right\} \tag{3}$$

where, $P_{EES,t}$ is the EES charging/discharging power at time t; fr_t is the frequency regulation power at time t; r_t is the frequency regulation revenue at time t. π_t is the electricity price at time t.

3.1.2. Constraints

The optimization problem is further restricted by the following constraints to ensure technically feasible operation:

3.1.2.1. EES charge/discharge power limits. EES can only be charged or discharged within the allowable range defined by its rated power. The total EES power used for charging/discharging and reserved for frequency regulation does not exceed the rated EES power.

$$-P_{EES}^{rated} \le P_{EES,t} \le P_{EES}^{rated} \quad \forall t = 1, \ldots, T \tag{4}$$

$$\left| P_{EES,t} \right| + fr_t \le P_{EES}^{rated} \; \forall t = 1, \ldots, T \tag{5}$$

$$P_{EES}^{rated} = E_{EES}^{rated} / \gamma \tag{6}$$

where, $P_{EES,t}$ is EES charging/discharging power at time t, where a positive value indicates charging and a negative value indicates discharging; P_{EES}^{rated} is the maximum charge/discharge rate (kW); E_{EES}^{rated} is the EES capacity (kWh). γ is the energy to power ratio (h); fr_t is the frequency regulation power at time t (kW).

3.1.2.2. PV-load balancing constraints. The energy balance constraint is fundamental to ensuring the operational feasibility of the integrated energy system. At each time step t, the total electricity supply must precisely meet the data center's total power demand. This constraint is mathematically expressed as:

$$P_{dem,t} = P_{PV,t}^{use} + P_{grid,t} + P_{EES,t} \quad \forall t = 1, \ldots, T \tag{7}$$

where, $P_{PV,t}^{use}$ denotes the photovoltaic power actively consumed by the data center at time t; $P_{grid,t}$ represents the power imported from the main grid at time t; $P_{EES,t}$ denotes the net discharge power from the energy storage system at time t.

EES discharge is limited not only by EES power but also by the net energy shortfall between PV generation and IT load:

$$\max\left(0, P_{dem,t} - P_{PV,t}^{gen}\right) \le P_{EES,t} \quad \forall t = 1, \ldots, T \tag{8}$$

where, $P_{dem,t}$ is the IT load and $P_{PV,t}^{gen}$ is PV output at time t.

3.1.2.3. State of charge (SOC) dynamics and bounds

$$SOC_{t+1} = SOC_t + \frac{P_{EES,t} \cdot \Delta t}{E_{ESS}^{rated}} \quad \forall t = 1, \ldots, T \tag{9}$$

$$SOC_{\min} \le SOC_t \le SOC_{\max} \quad \forall t = 1, \ldots, T+1 \tag{10}$$

The state-of-charge (SOC) of the EES is constrained within a safe operating window, typically between 20% and 80% of its nominal capacity [38,47]. This range is selected to prevent overcharging and deep discharging, both of which can significantly degrade battery performance and reduce its useable lifespan [48]. In addition, the maximum charging and discharging rates are determined based on the battery's power electronics and thermal limits. These constraints ensure safe and efficient operation and are typically expressed as a percentage of the rated energy capacity per hour.

3.1.2.4. Frequency regulation SOC margin. The following constraints are imposed to ensure sufficient SOC margin for frequency regulation operation by reserving both upward and downward SOC flexibility for symmetric regulation service. The factor of 0.5 assumes that, within each regulation interval, the battery may need to respond to approximately half of the committed regulation capacity in either charging or discharging direction.

$$SOC_{t+1} - \frac{0.5 \cdot fr_t \cdot \Delta t}{E_{EES}^{rated}} \ge SOC_{\min}, \forall t = 1, \ldots, T \tag{11}$$

$$SOC_{t+1} + \frac{0.5 \cdot fr_t \cdot \Delta t}{E_{EES}^{rated}} \le SOC_{\max}, \forall t = 1, \ldots, T \tag{12}$$

3.1.2.5. Initial and final SOC. The initial SOC is scenario-dependent and is set to balance surplus and deficit in PV generation. The operational rule for the EES system is to maximize the utilization of PV power. The system charges by absorbing excess PV generation and then discharges to return to its initial state of charge. During this discharge phase, the EES can be scheduled to provide flexibility services to the grid, generating revenue. This cyclical process then resets to begin a new cycle. The battery's operational depth of discharge is set to 60%, corresponding to a useable SOC range of 20%–80% of the full capacity, since operating beyond this range may accelerate battery degradation and shorten its lifespan. When the battery's operational depth of discharge is set at 60%, the losses incurred from participating in grid regulation services throughout the life cycle are nearly equivalent to the losses experienced when leaving the energy storage systems unused [38,47].

It is worth noting that the initial SOC is defined at the beginning of each optimization interval, corresponding to the transition point from PV surplus to deficit conditions. The final SOC is defined at the end of each optimization interval, corresponding to the completion of the PV

deficit period.

The initial SOC is determined based on the relative balance between PV surplus (A_{sur}) and PV deficit (A_{def}) within the window:

- Case 0: No PV Surplus ($A_{sur} = 0$): No PV surplus exists during the period; the EES should begin at minimum SOC (20%) since there is no opportunity to charge from PV.

$$\mathrm{SOC_{init}} = \mathrm{SOC_{min}} \tag{13}$$

- Case 1: If PV Surplus < PV Deficit ($A_{sur} < A_{def}$). Subcase 2.1: EES is large enough to shift all surplus to deficit periods, though not enough to meet total demand. Subcase 2.2: EES capacity is insufficient for both surplus absorption and full deficit compensation.

$$\mathrm{SOC_{init}} = \begin{cases} \mathrm{SOC_{min}} + \dfrac{A_{sur}}{E_{EES}^{rated}}, & \text{if } 0.6 \times E_{EES}^{rated} \geq A_{sur} \\ \mathrm{SOC_{max}}, & \text{otherwise} \end{cases} \tag{14}$$

- Case 2: PV Surplus ≥ PV Deficit ($A_{sur} \geq A_{def}$). Subcase 1.1: EES capacity is sufficient to absorb all surplus and cover the deficit. Subcase 1.2: EES capacity is insufficient to store all surplus energy.

$$\mathrm{SOC_{init}} = \begin{cases} \mathrm{SOC_{min}} + \dfrac{A_{def}}{E_{EES}^{rated}}, & \text{if } 0.6 \times E_{EES}^{rated} \geq A_{def} \\ \mathrm{SOC_{max}}, & \text{otherwise} \end{cases} \tag{15}$$

To maximize the utilization of surplus photovoltaic (PV) energy stored in the EES, the final state of charge (SOC) is fixed at 20% of the total storage capacity.

$$\mathrm{SOC_{final}} = \mathrm{SOC_{min}} \tag{16}$$

3.2. Performance indicators and economic assessment

To comprehensively evaluate the performance of the proposed PV-EES integrated energy system for data centers, both technical and economic metrics are considered. These include indicators related to renewable energy utilization, EES operation, system efficiency, and life-cycle economic performance. This section defines the key performance metrics and outlines the methodology for life-cycle cost analysis.

3.2.1. Renewable energy utilization

Renewable energy utilization quantifies the proportion of the data center's energy demand that is met by on-site PV generation. It reflects the effectiveness of the PV-EES system in reducing grid dependence and carbon emissions. This metric is also referred to as solar self-consumption and is calculated as:

$$RE_{uti} = \frac{\sum_{t=1}^{T} P_{PV,t}^{use}}{\sum_{t=1}^{T} P_{dem,t}} \tag{17}$$

where, RE_{uti} is the renewable energy utilization; $P_{dem,t}$ is the IT load and $P_{PV,t}^{use}$ is PV power consumed by the data center (either directly or via storage) at time t.

3.2.2. EES utilization and PV curtailment

EES utilization (EES_{uti}) reflects how effectively the available storage capacity is used to balance supply and demand. For example, if the EES is oversized while the PV capacity is relatively small, the EES utilization rate will be low, as a portion of the storage capacity will remain unused. The EES utilization is defined on a cycle basis, where each cycle represents an independent operational period characterized by its initial and final SOC.

$$EES_{uti} = \frac{1}{N_c} \sum_{c=1}^{N_c} \frac{SOC_{c,init} - SOC_{c,final}}{SOC_{\max} - SOC_{\min}} \tag{18}$$

where, N_c is the total number of cycles; $SOC_{c,init}$ and $SOC_{c,final}$ denote the initial and final state of charge of the EES in the cycle c, respectively.

PV curtailment refers to the portion of generated solar energy that is not utilized due to storage limitations or lack of demand. This is especially relevant in scenarios where PV surplus exceeds PV deficit, but the EES capacity is insufficient to absorb all the surplus. The curtailment rate is defined as:

$$P_{PV,t}^{cur} = P_{PV,t}^{gen} - P_{PV,t}^{use} \tag{19}$$

$$PVR_{cur} = \frac{\sum_{t=1}^{T} P_{PV,t}^{cur}}{\sum_{t=1}^{T} P_{PV,t}^{gen}} \tag{20}$$

where, $P_{PV,t}^{cur}$ is the PV curtailment power at time t (kW); PVR_{cur} is the PV curtailment rate.

3.2.3. Life-cycle cost analysis

To support investment decisions and system design optimization, a life-cycle cost analysis is conducted. Life-cycle cost provides a comprehensive financial evaluation by accounting for all relevant costs and revenues over the system's lifetime. The objective is to minimize the total life-cycle cost, which is given by:

$$C_{tot} = C_{inv} + C_{rep} + \sum_{i=1}^{I} \frac{(C_{O\&M,i} - Profit_i)}{(1+d)^i} \tag{21}$$

where, C_{tot} is the life-cycle cost, C_{inv} is the total investment cost throughout the lifetime, C_{rep} is the total replacement cost throughout the lifetime, $C_{O\&M,i}$ is the maintenance cost at ith year, $Profit_i$ is the total revenue at ith year, d is the discount rate. I is the financial timeline. i is the ith year after the investment.

The baseline cost (C_{base}) is the total cost of purchasing electricity from the grid without any energy storage intervention. Optimized cost is the amount of energy needed to meet the demand after considering the available PV power.

$$C_{base} = \sum_{t=1}^{T} \pi_t \times P_{dem,t} \times \Delta t \tag{22}$$

3.3. Framework for optimal dispatch and design of PV-EES systems

This section introduces a comprehensive optimization framework for identifying the optimal PV and EES capacity configurations in data centers. The framework combines technical performance analysis, operational optimization, and lifecycle economic assessment to support decision-making under different design objectives. Fig. 4 provides an overview of the proposed methodology. The overall optimization process consists of four primary steps:

i. The first step is to define the baseline characteristics of the data center, including the design IT load profile, representative PV system configurations and electrical energy storage system. Key input data are collected, including progressive IT load profiles, typical solar irradiance data with hourly resolution, and energy prices including grid electricity tariffs and flexibility market revenue.
ii. The optimization process is conducted in two sequential stages. The first-step optimization determines the optimal hourly dispatch strategy for a given PV-EES system, such as hourly charging/discharging and flexibility services (e.g., frequency

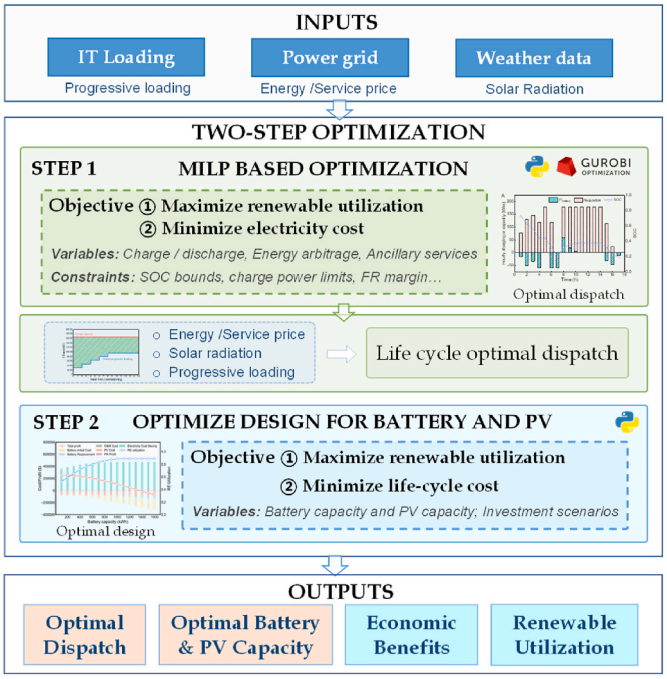


Fig. 4. Schematic of multi-step optimization for PV-EES integrated data centers.

regulation). The optimization problem is formulated to maximize renewable energy utilization while minimizing operational costs on a daily basis.

iii. The second-step optimization identifies the optimal capacity combination of PV and EES systems for the reference data center, considering lifecycle progressive loading. Based on the daily dispatch algorithm, annual dispatch profiles can be obtained for specific loading conditions and locations. Subsequently, lifecycle performance metrics under various PV-EES configurations, such as renewable energy utilization, lifecycle net profit, grid reliance and PV curtailment, can be obtained.

iv. The optimal PV-EES capacity combinations are further examined under different decision priorities. For example, if the design priority is to maximize renewable energy utilization, larger PV arrays and moderate battery sizes may be preferred to capture and use as much solar energy as possible.

4. Optimal dispatch results of electrical energy storage system

This section presents hourly-resolution optimal dispatch strategies of EES systems under two representative regional electricity markets: Shenzhen, China, and CAISO, United States. This section analyzes representative days selected from each loading stage across the data center lifecycle. The base configuration of the reference data center includes a battery storage system with a capacity of 500 kWh and a photovoltaic generation system with an installed capacity of 150 kW.

4.1. Results under Shenzhen Electricity Market in China

Fig. 5 shows the IT load during stage 5, which represents the longest operating period in the data center lifecycle, together with the electricity pricing signals (including both energy prices and ancillary service revenue) and solar irradiance data for a representative day in Shenzhen, China (January 1, 2023). In Fig. 5A, the shaded areas represent the periods when the EES participates in charging, discharging, or grid-support services such as frequency regulation. These operating periods are identified dynamically according to PV surplus and deficit conditions described in Fig. 3, and they change with the real-time variations in IT demand and solar power generation.

Fig. 5B illustrates the resulting EES dispatch strategy during the selected operating periods. During most of these intervals, the EES operates in discharge mode, while a greater share of the EES capacity is assigned to frequency regulation instead of energy arbitrage. This behavior is mainly driven by operational limitations and the requirement to release the renewable electricity stored earlier in the same day. Throughout the dispatch process, the SOC remains within the allowable operating range, helping maintain EES safety and long-term durability.

Fig. 6 further illustrates how the EES operating strategy changes throughout different stages of the data center lifecycle. Although the PV generation profile remains the same, the EES dispatch behavior differs considerably among the four stages. The main reason is the evolution of the IT load profile over time, which changes the timing and duration of PV surplus and deficit periods. As a result, the intersection between the IT load and PV generation curves also shifts (Fig. 3).

During the earlier operating stages (Stage 1 and Stage 2), the EES mainly performs discharge-only operation because of the relatively low IT load and the stricter end-of-day SOC requirements. In contrast, both charging and discharging behaviors appear in the later stages (Stage 3 and Stage 4). This bidirectional operation is associated with the larger variation in IT load, which provides more flexibility for coordinating EES dispatch with the net system load while still satisfying the SOC limits and final SOC constraints.

4.2. Results under CAISO electricity market in the United States

Fig. 7 shows the IT load, solar irradiance, and electricity pricing data, including both energy prices and ancillary service revenues, for a representative day (January 1, 2023) in San Francisco, United States. In contrast to the relatively regulated or flatter pricing structure in Shenzhen, the CAISO market adopts an hourly day-ahead pricing

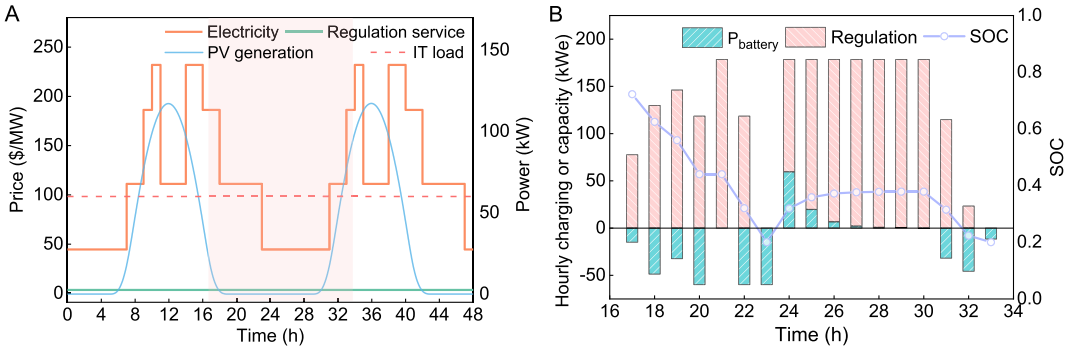


Fig. 5. IT load profile, electricity pricing and solar irradiance on two representative days in Shenzhen (A); Optimal dispatch results of electrical energy storage system at stage 5 (60% IT load) of data center lifecycle (B).

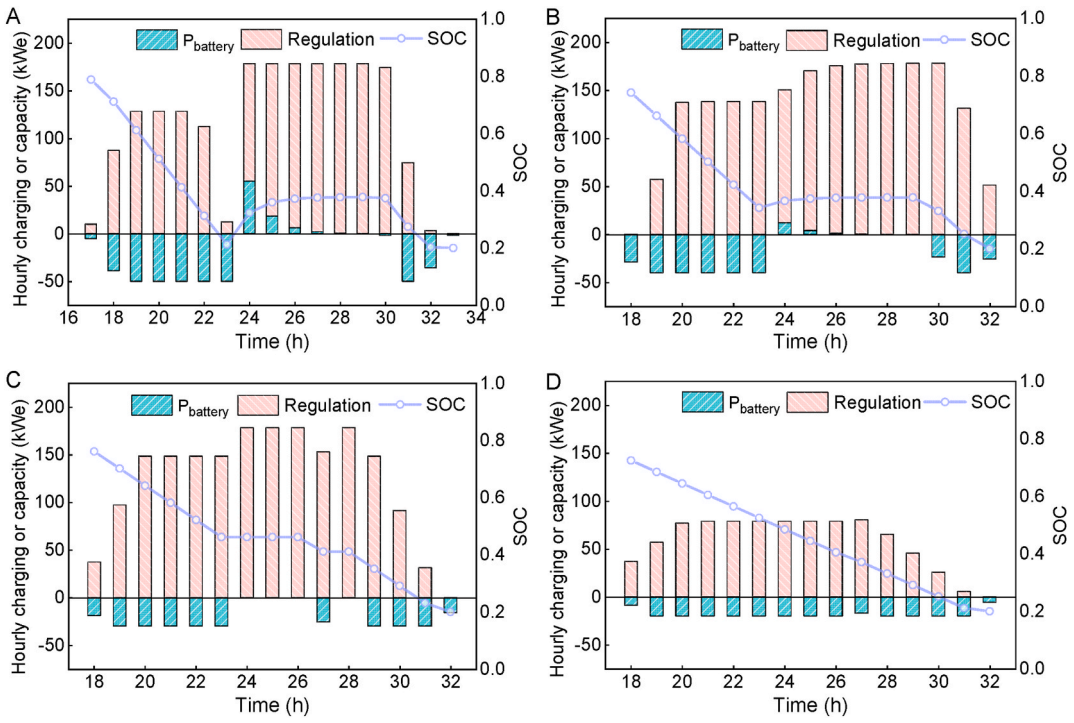


Fig. 6. Optimal dispatch results of electrical energy storage system on a representative day in Shenzhen across different stages, (A) Stage 4 (50% IT load); (B) Stage 3 (40% IT load); (C) Stage 2 (30% IT load); (D) Stage 1 (20% IT load).

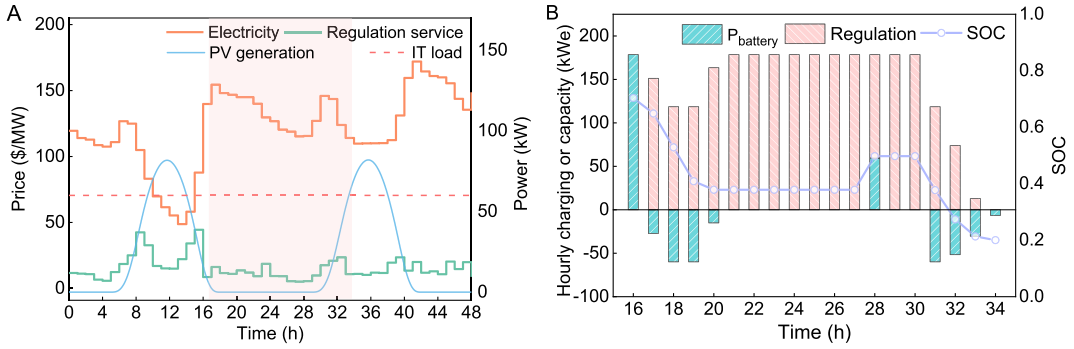


Fig. 7. (A) IT load profile, electricity pricing and solar irradiance on a representative day in San Francisco; (B) Optimal dispatch results of the electrical energy storage system at stage 5 (60% IT load) of the data center lifecycle.

mechanism with continuously varying electricity prices. This difference in market characteristics results in a different operating strategy for the EES. As in the previous figures, the shaded areas in Fig. 7A represent the time periods when the EES participates in grid-support activities, including energy arbitrage through charging and discharging, as well as ancillary services such as frequency regulation.

Fig. 7B further shows the optimal EES dispatch profile during these active periods for Stage 5 of the data center lifecycle. The results indicate that the EES charges during the initial hour and then switches to discharge operation in the following hours. This operating pattern is determined by the combined effects of the initial SOC, IT load conditions, solar generation availability, and hourly electricity prices on that day. The early charging behavior is mainly associated with storing low-cost electricity in advance so that the EES can later discharge during higher-price periods for energy arbitrage benefits. At the same time, part of the EES capacity is reserved to support participation in the frequency regulation market, providing additional economic value.

Fig. 8 further compares the EES operating patterns at different stages of the data center lifecycle. The dispatch strategy observed in Stage 4 generally follows the same trend as Stage 5, where the EES charges during low-price periods in the early hours and discharges when electricity prices increase. In contrast, during Stages 2 to 4, the lower IT load reduces the profitability of energy arbitrage. As a result, charging activity becomes much less frequent, and a larger share of the EES capacity is used for ancillary services, particularly frequency regulation. Across all lifecycle stages, the SOC profile consistently stays within the specified operational boundaries.

5. Analysis on lifecycle profitability and renewable utilization

This section summarizes the optimal sizing results for the PV and EES systems. Using the annual hourly dispatch simulations, key performance indicators, including annual electricity cost reduction and revenues from frequency regulation services, are calculated. The analysis mainly evaluates lifecycle economic performance, renewable energy utilization, and the trade-offs between these objectives under different PV–EES sizing combinations.

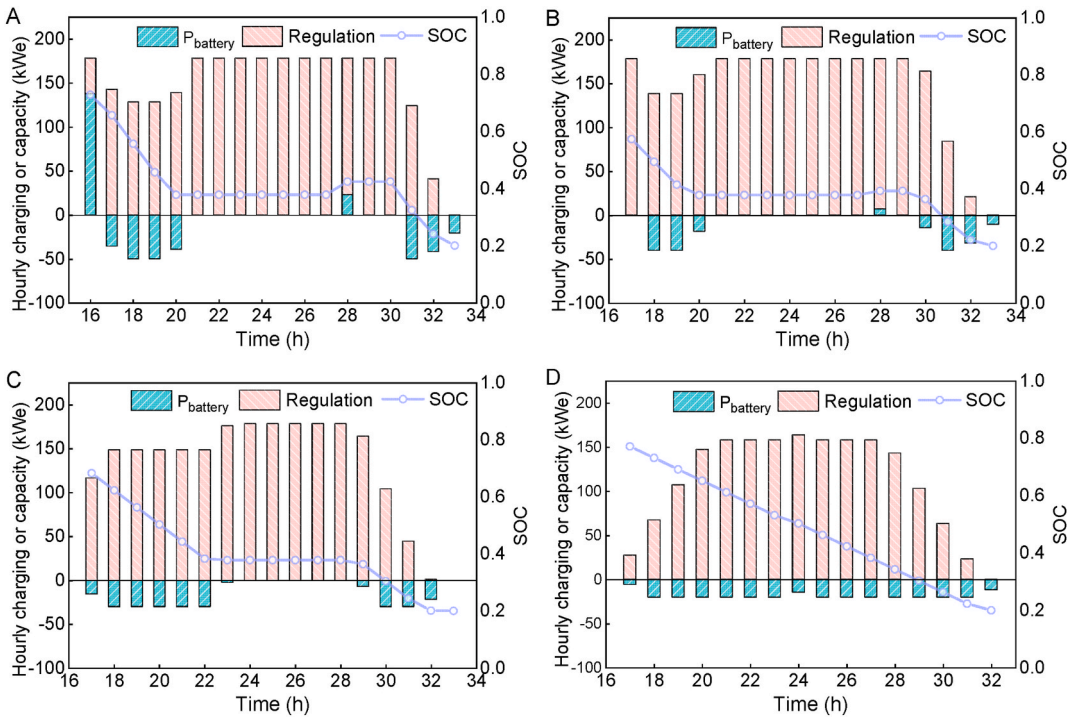


Fig. 8. Optimal dispatch results of electrical energy storage on a representative day in San Francisco across different stages, (A) Stage 4 (50% IT load); (B) Stage 3 (40% IT load); (C) Stage 2 (30% IT load); (D) Stage 1 (20% IT load).

5.1. Impact of EES capacity on system economic performance

Fig. 9 compares the total lifecycle costs and profits for different EES capacities, ranging from 100 kWh to 1600 kWh, while maintaining a constant PV capacity of 150 kW under both the Shenzhen and CAISO electricity markets. The evaluated economic terms include PV installation cost, EES investment and replacement costs, operation and maintenance expenses, electricity cost savings, and revenues from frequency regulation services.

For the Shenzhen market (Fig. 9A), the overall net profit gradually decreases as EES capacity becomes larger, changing from a highly positive value to a negative one at high capacities. The net profit drops from about $253,100 at 100 kWh to -$71,026 at 1600 kWh. This reduction is mainly caused by the increasing EES purchase, replacement, and O&M costs associated with larger storage systems. At the same time, larger batteries generate higher electricity cost savings and greater frequency regulation revenues, which partially compensate for the increased expenses.

Fig. 9B shows a comparable pattern under the CAISO market. The net profit declines from $31,513 at 100 kWh to -$96,375 at 1600 kWh as EES capacity increases. Compared with Shenzhen, the CAISO market produces lower profitability across the entire EES capacity range. The main reason is that the electricity cost savings in CAISO are smaller, even though the frequency regulation revenues are slightly higher under these market conditions.

The renewable energy utilization results show an increasing trend under both electricity markets, reflecting the fraction of PV-generated electricity that can be directly consumed or stored by the system. In both cases, the improvement gradually slows once EES capacity exceeds around 1000 kWh, suggesting limited additional gains in PV self-consumption at larger storage sizes. However, the net profit continues to decline with further EES expansion, indicating that the economic benefit of adding more storage becomes progressively weaker.

In the Shenzhen market, renewable energy utilization rises continuously from 52.5% at 100 kWh to 88.8% at 1000 kWh. A similar trend can also be observed in the CAISO market, although the utilization levels remain slightly lower, reaching 80.9% at 1000 kWh. These differences may be related to variations in solar resource conditions, grid operation characteristics, and EES dispatch optimization strategies between the two regions.

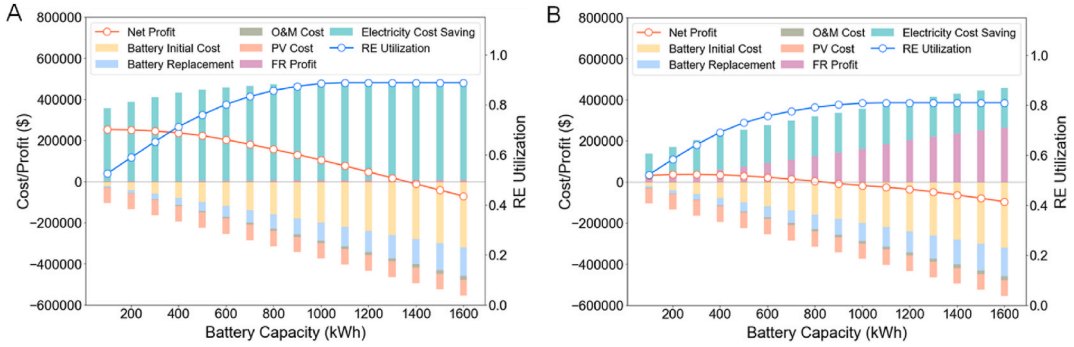


Fig. 9. System economics and renewable utilization as a function of EES capacity for a 150 kW PV system. (A) Shenzhen electricity market; (B) CAISO electricity market.

The results demonstrate a clear trade-off between economic performance and renewable energy utilization. Increasing EES capacity can improve renewable energy self-consumption, but it also introduces higher capital investment costs. In the Shenzhen market, the system can still maintain positive net profit even when renewable utilization approaches its highest level. In contrast, under the CAISO market, achieving high renewable energy utilization is associated with negative net profit.

5.2. Impact of PV capacity on system economic performance

Fig. 10 compares the total lifecycle costs and profits under different PV capacities, ranging from 30 kW to 510 kW, for both the Shenzhen and CAISO electricity markets while keeping the EES capacity fixed at 500 kWh. The stacked bars show the contribution of each economic component, including PV installation cost, EES investment and replacement costs, O&M expenses, electricity cost savings, and revenues from frequency regulation services.

For the Shenzhen market (Fig. 10A), the net profit first increases as PV capacity becomes larger and reaches its maximum value of $225,919 at 120 kW. The improvement in profitability is mainly associated with higher PV self-consumption, which reduces electricity purchases from the grid. However, after this turning point, the increase in profit gradually slows and eventually declines at higher PV capacities. This trend is mainly caused by the limited battery storage capacity, which constrains the system's ability to utilize additional PV generation effectively. As a result, the economic benefit gained from further PV expansion becomes progressively smaller.

The renewable energy utilization, defined as the fraction of PV-generated electricity relative to the total system energy consumption, increases steadily from 0.23 at 30 kW to 0.81 at 510 kW, indicating improved renewable energy integration with larger PV systems. Nevertheless, the growth rate becomes slower after the PV capacity reaches approximately 120 kW. This slower increase is mainly related to the limited battery storage capacity, which reduces the system's ability to store and effectively use additional PV generation.

A comparable trend can also be observed in the CAISO market (Fig. 10B) for both renewable energy utilization and net profit. The maximum net profit appears at a PV capacity of 120 kW, reaching $36,552. After this point, profitability gradually decreases, and the system begins to show negative net profit once the PV capacity reaches 220 kW. This reduction is mainly caused by the higher PV investment cost, which cannot be fully compensated by electricity cost savings and frequency regulation revenues at larger PV capacities. The renewable energy utilization trend under the CAISO market is generally similar to that observed in Shenzhen.

Overall, both electricity markets exhibit the same general relationship between PV capacity, profitability, and renewable energy utilization. Net profit first rises and then declines as PV capacity increases, with the optimal economic point occurring near 120 kW in both cases. Renewable energy utilization also increases continuously with larger PV capacities in both markets. However, the Shenzhen market achieves better economic performance overall. This advantage mainly results from greater electricity cost savings in Shenzhen, whereas the CAISO market provides comparatively lower electricity savings despite slightly higher revenues from frequency regulation services.

5.3. Trade-off between lifecycle profitability and renewable utilization

5.3.1. Profit-optimal configurations

Fig. 11 presents the net profit results for different combinations of EES capacity (100–1600 kWh) and PV capacity (30–510 kW) under the Shenzhen and CAISO electricity markets. The results reveal clear differences in the profitability distributions between the two market environments.

In the Shenzhen market (Fig. 11A), a relatively large range of PV–EES combinations achieves positive net profit. The highest profitability appears near a PV capacity of 120 kW combined with EES capacities between 100 and 300 kWh. In contrast, systems with very large EES capacities but relatively small PV systems experience substantial financial losses exceeding $-300,000, indicating excessive investment in storage capacity without sufficient renewable generation to support it.

The profitability pattern under the CAISO market (Fig. 11B) is considerably less favorable. Most PV–EES combinations produce negative net profit, and only a limited region shows slightly positive economic performance. Even under the most favorable conditions, such as EES capacities between 100 and 500 kWh with PV capacities close to 120 kW, the net profit generally remains below $40,000 and decreases rapidly outside this range.

The large difference between the two markets is mainly related to variations in electricity pricing structures and market incentives. Compared with CAISO, the Shenzhen market provides stronger economic support for integrated PV and EES systems, likely because of larger time-of-use electricity price differences. These results highlight the importance of conducting location-specific techno-economic analysis when determining PV and EES capacities. Excessively large PV or storage systems may significantly reduce profitability if local market conditions are not properly considered. Therefore, the optimal design should balance renewable generation and storage capacity according to regional electricity market characteristics.

5.3.2. Renewable-optimal configurations

Fig. 12 shows the renewable energy utilization of hybrid PV–EES systems under different sizing combinations, with PV capacities varying from 30 kW to 510 kW and EES capacities ranging between 100 kWh and 1600 kWh. The values in each heatmap cell represent the fraction of renewable energy effectively utilized by the system, covering a range from about 0.2 to nearly 1.0.

For the Shenzhen market (Fig. 12A), renewable energy utilization gradually increases as both PV and EES capacities become larger.

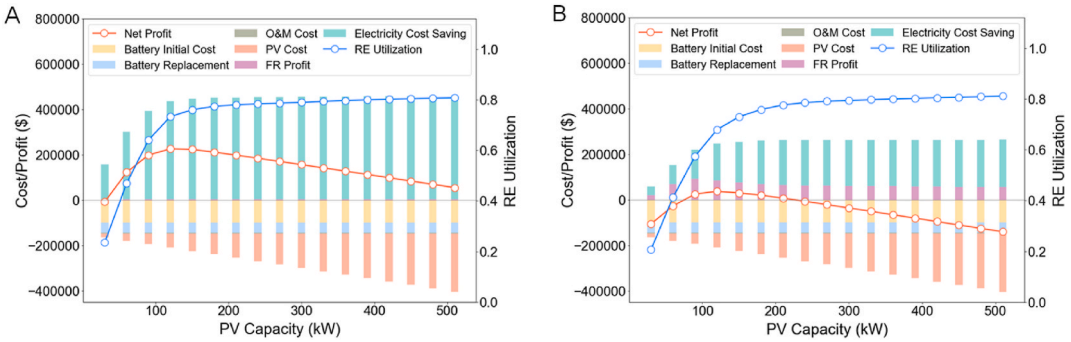


Fig. 10. System economics and renewable utilization as a function of PV capacity for a 500 kWh EES system. (A) Shenzhen electricity market; (B) CAISO electricity market.

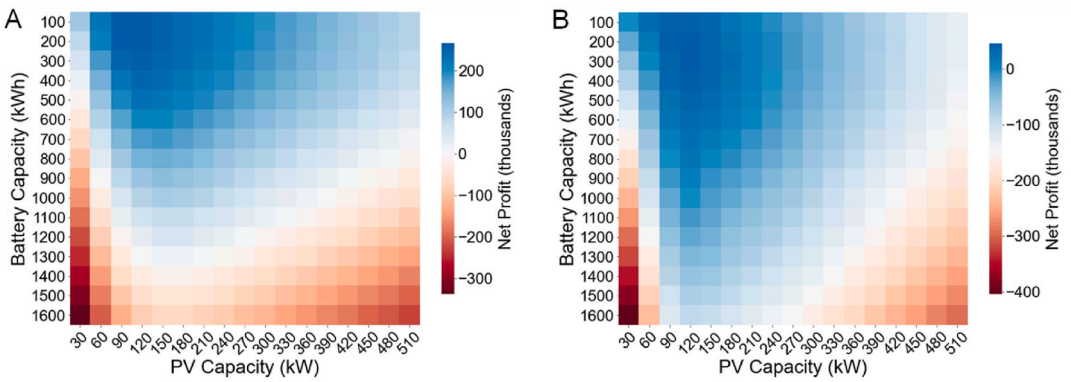


Fig. 11. Lifecycle net profit across different PV-EES configurations under the (A) Shenzhen Electricity Market and (B) CAISO Electricity Market.

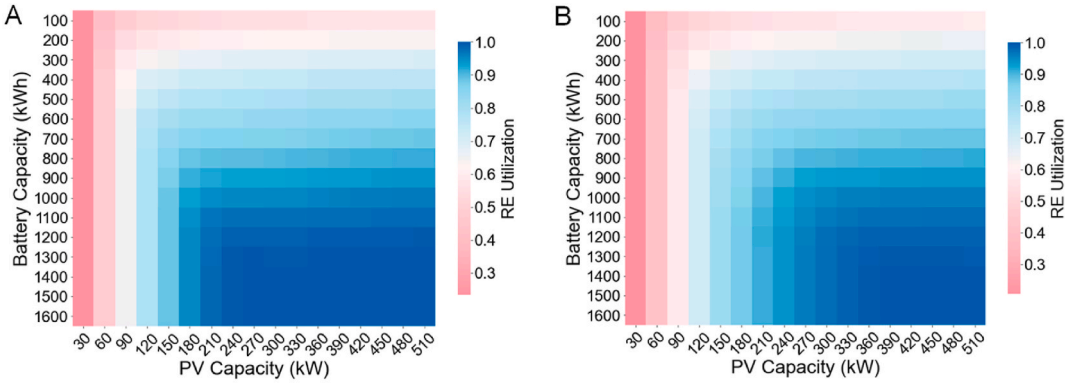


Fig. 12. Renewable energy utilization across different PV-EES configurations under the (A) Shenzhen electricity market and (B) CAISO electricity market.

Systems with EES capacities above 1000 kWh and PV capacities exceeding 270 kW can consistently achieve utilization levels higher than 90%. Under the CAISO market (Fig. 12B), the renewable energy utilization follows a similar overall trend but remains slightly lower, especially for medium-scale system configurations. For example, the smallest system configuration, consisting of a 100 kWh EES and 30 kW PV system, achieves a utilization of approximately 20.7%, which is very close to the Shenzhen result. However, as the PV and EES capacities increase, the renewable energy utilization in CAISO becomes somewhat lower than that in Shenzhen. This difference is mainly related to variations in solar radiation conditions and IT load characteristics, which influence the EES dispatch and energy management strategy.

5.3.3. *In-depth trade-off analysis*

Although improving renewable energy utilization is an important goal for sustainable energy systems, the configurations that maximize renewable usage are not always the most economically favorable. As shown in Sections 5.3.1 and 5.3.2, the highest renewable energy utilization levels generally do not coincide with the highest net profits, revealing a clear trade-off between economic and environmental objectives.

In the Shenzhen market, the highest net profit, exceeding $200,000, is achieved with moderate EES capacities of 100–300 kWh and PV capacities near 120 kW. However, these economically optimal configurations only reach renewable energy utilization levels of about 40%–65%. In contrast, systems with battery capacities larger than 1000 kWh and PV capacities above 270 kW can achieve renewable energy utilization greater than 95%, but the associated profits decrease substantially and may even become negative.

A similar trade-off is observed in the CAISO market, although the overall profitability is much lower. Positive economic performance is limited to a relatively small range of configurations, typically involving EES capacities of 100–500 kWh and PV capacities between 90 and 150 kW, where the net profit rarely exceeds $40,000. These cases are associated with renewable energy utilization levels of roughly 35%–60%. Increasing PV and EES capacities can further improve renewable energy utilization, exceeding 90% for systems such as a 1500 kWh battery combined with a 390 kW PV system. However, these large-scale configurations also produce net losses greater than $-250,000. This result suggests that the CAISO market structure provides limited economic benefit for surplus renewable generation and large storage capacities, reducing the financial attractiveness of high-renewable-utilization designs.

These results demonstrate an important system design challenge. Configurations targeting very high renewable energy utilization usually require larger PV and EES investments, leading to longer payback periods and reduced economic performance. This issue becomes more significant in electricity markets with relatively weak pricing incentives. Therefore, the design of PV-plus-storage systems should consider both profitability and renewable energy utilization within a multi-objective optimization framework.

5.4. *Impact on EES utilization and PV curtailment*

5.4.1. *EES utilization under different configurations*

Fig. 13 presents the EES utilization rates for different combinations of EES capacity (100–1600 kWh) and PV capacity (30–510 kW) under the Shenzhen and CAISO electricity markets. The heatmaps show noticeable differences in EES utilization behavior between the two market conditions.

For both markets, EES utilization generally increases as PV capacity becomes larger, especially when the EES size is relatively small. In the Shenzhen market (Fig. 13A), for instance, a 100 kWh EES paired with a 30 kW PV system achieves a utilization of only 9.0%, whereas the

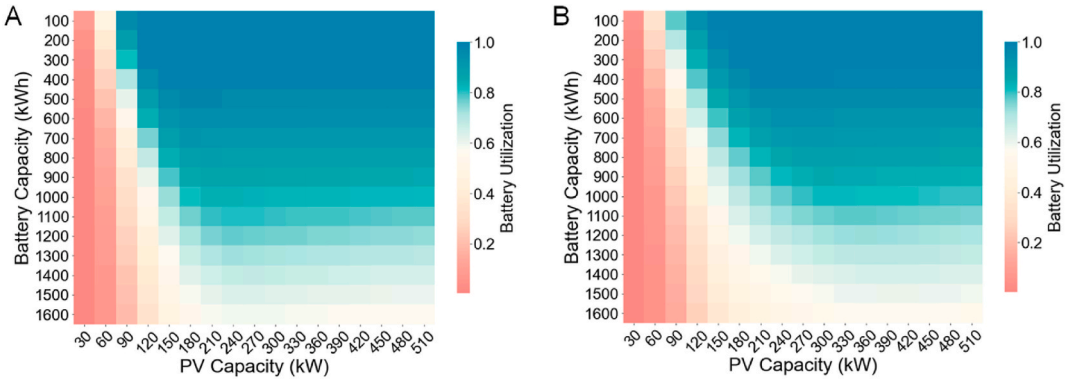


Fig. 13. Heatmaps of EES utilization rate across different PV-EES Configurations for (A) Shenzhen electricity market and (B) CAISO electricity market.

utilization rises to 100% once the PV capacity reaches 120 kW or above. The CAISO market (Fig. 13B) shows a comparable trend, although the utilization values are usually lower under the same PV–EES combinations.

At the same time, the benefit of increasing PV capacity becomes less significant when EES capacity is very large. For example, in Shenzhen, a 1600 kWh EES combined with a fixed 90 kW PV system achieves a utilization rate of only 5.6%, indicating that a large portion of the storage capacity remains unused.

Comparing the two markets, the Shenzhen case generally achieves higher EES utilization for the same system configurations. This difference is likely related to variations in solar radiation conditions and IT load characteristics, which influence EES dispatch behavior and storage operation. Overall, the results emphasize the importance of jointly optimizing PV and EES capacities to improve both utilization and economic performance. Small PV systems may not provide enough energy to effectively charge the EES, while excessively large batteries can become costly underutilized assets. The analysis also demonstrates that the optimal PV–EES configuration strongly depends on local market conditions, highlighting the need for region-specific techno-economic assessment.

5.4.2. PV curtailment under different configurations

Fig. 14 shows the PV curtailment rates under different combinations of EES capacities (100–1600 kWh) and PV capacities (30–510 kW) for the Shenzhen and CAISO electricity markets. The heatmaps indicate clear differences in renewable energy curtailment behavior between the two market scenarios.

In the Shenzhen market (Fig. 14A), PV curtailment is relatively high when EES capacity is small, especially for systems with large PV installations. For example, with a 100 kWh EES and a 510 kW PV system, the curtailment rate reaches about 83.7%, meaning that a large fraction of the generated solar electricity cannot be utilized because of limited storage capacity. As the EES size increases, the curtailment rate decreases substantially.

Under the CAISO market conditions (Fig. 14B), the curtailment levels are generally lower for the same PV and EES combinations. At the same configuration of 510 kW PV and 100 kWh storage, the curtailment rate is about 81.2%, which is slightly below the Shenzhen result.

Comparing the two markets, Shenzhen tends to experience higher PV curtailment under configurations with large PV capacity and limited EES storage, whereas the CAISO market achieves slightly better utilization of solar generation in most cases. For example, when the EES capacity is 600 kWh and the PV capacity is 300 kW, the curtailment rate in Shenzhen is around 60.8%, while the corresponding value in CAISO is approximately 55.4%. Overall, the results highlight the important role of EES sizing in reducing PV energy curtailment and improving renewable energy utilization.

6. Discussion and conclusions

This study developed a comprehensive optimization framework for the joint optimization of both system design and operation of onsite PV and EES under lifecycle progressive loading and participation in the smart grid. By integrating both technical and economic considerations under realistic operating conditions, this work demonstrates that large-scale carbon reduction in data centers can be achieved while maintaining favorable economic performance. The major findings of this study are summarized below.

Despite the considerable upfront investment required for onsite PV and EES deployment, grid-interactive data centers can still achieve high renewable energy penetration, significant carbon emission reductions, and strong lifecycle economic benefits. For example, a 100 kW data center integrated with 120 kW of PV capacity and 100 kWh of EES achieved renewable utilization rates of 50–51% in both electricity markets. Under these configurations, the lifecycle net profit reached

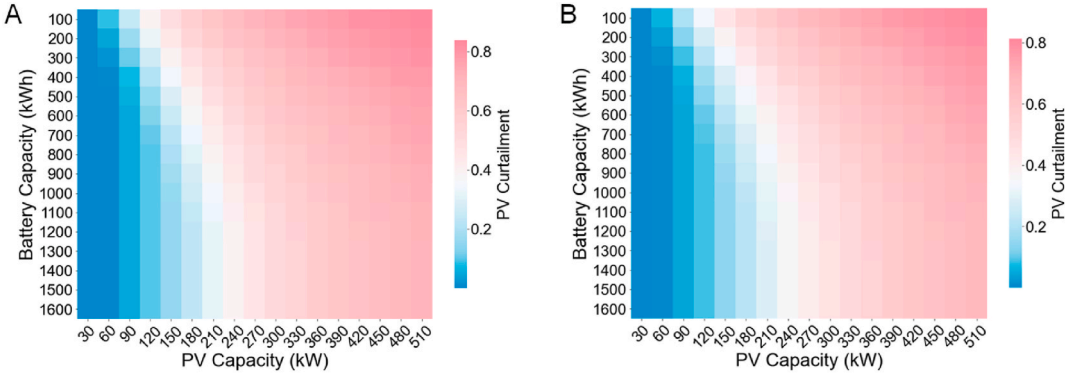


Fig. 14. Heatmaps of PV Curtailment Rate Across different PV-EES configurations for (A) Shenzhen electricity market and (B) CAISO electricity market.

$0.26 million in the Shenzhen market and $0.04 million in the CAISO market.

However, reaching renewable utilization levels above 90% while simultaneously maximizing lifecycle net profit requires substantial increases in both PV and EES capacities. In the Shenzhen market, this target requires 2.2 times more EES capacity and 8 times more PV capacity, while still maintaining a lifecycle net profit of $0.04 million. In the CAISO market, achieving the same renewable utilization level requires 1.4 times more EES capacity and 10 times more PV capacity, although the lifecycle net profit becomes negative under this condition. These results reveal a clear trade-off between maximizing renewable energy utilization and maximizing lifecycle economic performance. Configurations with the highest profitability generally did not achieve the highest renewable utilization rates, whereas systems designed for maximum renewable utilization often showed lower economic returns.

Several important directions can be explored in future research. First, additional economic benefits may be achieved by investigating second-life battery applications and phased investment strategies, which could reduce initial capital costs and improve financial performance. Second, introducing workload scheduling flexibility may help decrease dependence on EES, allowing smaller battery capacities while still maintaining high renewable energy utilization. Third, future studies should further investigate real-world deployment barriers, including site-specific space constraints, battery aging effects, investment uncertainty, and grid interconnection limitations. Addressing these factors is essential for accurately evaluating the long-term feasibility and scalability of highly renewable grid-interactive data centers.

CRediT authorship contribution statement

Yingbo Zhang: Writing – original draft, Software, Methodology, Investigation, Data curation, Conceptualization. **Hong Tang:** Methodology, Investigation, Data curation. **Zilong Zhao:** Methodology, Formal analysis. **Xin Hu:** Writing – review & editing, Methodology, Investigation. **Zixuan Kang:** Writing – review & editing, Software, Methodology, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

I have shared the link of the data.

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